Day 6-Assignment

Set 1 – Retail Sales Reporting

1.1 Top 5 products by revenue

```
SELECT TOP (5) p.ProductName, SUM(oi.Quantity * oi.UnitPrice) AS Revenue
FROM dbo.OrderItems oi JOIN dbo.Products p ON p.ProductID = oi.ProductID
GROUP BY p.ProductID, p.ProductName ORDER BY Revenue DESC;
```

1.2 Month-wise sales

```
SELECT CONVERT(char(7), o.OrderDate, 120) AS SalesMonth, SUM(oi.Quantity * oi.UnitPrice) AS Revenue
FROM dbo.Orders o JOIN dbo.OrderItems oi ON oi.OrderID = o.OrderID
GROUP BY CONVERT(char(7), o.OrderDate, 120) ORDER BY SalesMonth;
```

1.3 Customers with more than 3 orders (HAVING)

```
SELECT CONCAT(c.FirstName, ' ', c.LastName) AS Customer, COUNT(*) AS OrderCount
FROM dbo.Customers c JOIN dbo.Orders o ON o.CustomerID = c.CustomerID
GROUP BY c.CustomerID, c.FirstName, c.LastName HAVING COUNT(*) > 3 ORDER BY OrderCount DESC;
```

1.4 Average basket value by city

```
WITH B AS (SELECT o.OrderID, c.City, SUM(oi.Quantity * oi.UnitPrice) AS BasketValue
        FROM dbo.Orders o JOIN dbo.Customers c ON c.CustomerID = o.CustomerID
        JOIN dbo.OrderItems oi ON oi.OrderID = o.OrderID GROUP BY o.OrderID, c.City)
SELECT City, CAST(AVG(BasketValue) AS decimal(10,2)) AS AvgBasketValue FROM B GROUP BY City ORDER
BY City;
```

Output for 1.1–1.4

```
ubuntu@sandbox:~$ /opt/mssql-tools18/bin/sqlcmd -C -S 127.0.0.1 -U sa -d RetailSalesReportDemo -i /home/ubuntu/retail_sales_set/q1a.sql -b -W -s "|"
Changed database context to 'RetailSalesReportDemo'.
1.1 Top 5 products by revenue
ProductName|Revenue
-----------|-------
Laptop|3200.00
Standing Desk|1680.00
Desk Lamp|300.00
Wireless Mouse|125.00
Ceramic Mug|108.00
1.2 Month-wise sales
SalesMonth|Revenue
----------|-------
2025-01|2915.00
2025-02|1436.00
2025-03|1142.00
1.3 Customers with more than 3 orders (HAVING)
Customer|OrderCount
--------|----------
Alice Chen|4
1.4 Average basket value by city
City|AvgBasketValue
----|--------------
Austin|459.00
Boston|225.33
Chicago|8.00
Seattle|572.00
ubuntu@sandbox:~$
```

1.5 Orders without payment (LEFT JOIN)

```
SELECT o.OrderID, CONCAT(c.FirstName, ' ', c.LastName) AS Customer, o.OrderDate
FROM dbo.Orders o JOIN dbo.Customers c ON c.CustomerID = o.CustomerID
LEFT JOIN dbo.Payments p ON p.OrderID = o.OrderID WHERE p.PaymentID IS NULL ORDER BY o.OrderID;
```

1.6 Customers with no orders

```
SELECT c.CustomerID, CONCAT(c.FirstName, ' ', c.LastName) AS Customer
FROM dbo.Customers c LEFT JOIN dbo.Orders o ON o.CustomerID = c.CustomerID
WHERE o.OrderID IS NULL ORDER BY c.CustomerID;
```

1.7 Product-category revenue share

```
WITH C AS (SELECT p.CategoryName, SUM(oi.Quantity * oi.UnitPrice) AS Revenue
        FROM dbo.OrderItems oi JOIN dbo.Products p ON p.ProductID = oi.ProductID GROUP BY p.CategoryName)
SELECT CategoryName, Revenue, CAST(100.0 * Revenue / SUM(Revenue) OVER() AS decimal(6,2)) AS
RevenueSharePct
FROM C ORDER BY Revenue DESC;
```

1.8 Day with the highest sales

```
WITH D AS (SELECT o.OrderDate, SUM(oi.Quantity * oi.UnitPrice) AS Revenue
        FROM dbo.Orders o JOIN dbo.OrderItems oi ON oi.OrderID = o.OrderID GROUP BY o.OrderDate)
SELECT TOP (1) OrderDate, Revenue FROM D ORDER BY Revenue DESC, OrderDate;
```

Output for 1.5–1.8

```
ubuntu@sandbox:~$ /opt/mssql-tools18/bin/sqlcmd -C -S 127.0.0.1 -U sa -d RetailSalesReportDemo -i /home/ubuntu/retail_sales_set/q1b.sql -b -W -s "|"
Changed database context to 'RetailSalesReportDemo'.
1.5 Orders without payment (LEFT JOIN)
OrderID|Customer|OrderDate
-------|--------|---------
5004|Alice Chen|2025-03-02
5008|Cara Singh|2025-03-18
5013|Hana Kim|2025-01-08
1.6 Customers with no orders
CustomerID|Customer
----------|--------
6|Farah Ali
7|Gideon Ross
1.7 Category revenue share
CategoryName|Revenue|RevenueSharePct
------------|-------|--------------
Electronics|3325.00|60.53
Furniture|1680.00|30.58
Home|408.00|7.43
Stationery|80.00|1.46
1.8 Day with the highest sales
OrderDate|Revenue
---------|-------
2025-02-10|860.00
ubuntu@sandbox:~$
```

2. Customer-friendly functions

```
SELECT TOP (5) UPPER(CONCAT(c.FirstName,' ',c.LastName)) AS Customer,
 FORMAT(o.OrderDate,'MMM dd, yyyy') AS OrderDate,
 DATEDIFF(day,o.OrderDate,CONVERT(date,GETDATE())) AS DaysSinceOrder,
 CONCAT('$',FORMAT(ROUND(SUM(oi.Quantity*oi.UnitPrice),2),'N2')) AS OrderTotal
FROM dbo.Orders o JOIN dbo.Customers c ON c.CustomerID=o.CustomerID
JOIN dbo.OrderItems oi ON oi.OrderID=o.OrderID
GROUP BY c.FirstName,c.LastName,o.OrderID,o.OrderDate ORDER BY o.OrderDate DESC,o.OrderID DESC;
```

Output

```
ubuntu@sandbox:~$ /opt/mssql-tools18/bin/sqlcmd -C -S 127.0.0.1 -U sa -d RetailSalesReportDemo -i /home/ubuntu/retail_sales_set/q2.sql -b -W -s "|"
Changed database context to 'RetailSalesReportDemo'.
Customer|OrderDate|DaysSinceOrder|OrderTotal
--------|---------|--------------|----------
EVA MARTIN|Mar 25, 2025|597|$132.00
DIEGO PARK|Mar 19, 2025|603|$480.00
CARA SINGH|Mar 18, 2025|604|$40.00
BEN ORTIZ|Mar 18, 2025|604|$445.00
ALICE CHEN|Mar 02, 2025|620|$45.00
ubuntu@sandbox:~$
```

3. NULL behavior

```
SELECT COUNT(*) AS AllItems,COUNT(DiscountPct) AS ItemsWithDiscount,
 CAST(AVG(DiscountPct) AS decimal(6,4)) AS AvgDiscountPct
FROM dbo.OrderItems;
```

Output

```
ubuntu@sandbox:~$ /opt/mssql-tools18/bin/sqlcmd -C -S 127.0.0.1 -U sa -d RetailSalesReportDemo -i /home/ubuntu/retail_sales_set/q3.sql -b -W -s "|"
Changed database context to 'RetailSalesReportDemo'.
AllItems|ItemsWithDiscount|AvgDiscountPct
--------|-----------------|--------------
22|9|.0722
ubuntu@sandbox:~$
```

4. Expected results

Query	Expected result
1.1 Top 5	Laptop $3,200; Standing Desk $1,680; Desk Lamp $300; Wireless Mouse $125; Ceramic Mug $108.
1.2 Monthly	Jan $2,915; Feb $1,436; Mar $1,142.
1.3 >3 orders	Alice Chen — 4 orders.
1.4 City basket	Austin $459.00; Boston $225.33; Chicago $8.00; Seattle $572.00.
1.5 Unpaid	Orders 5004, 5008, 5013.
1.6 No orders	Farah Ali; Gideon Ross.
1.7 Category share	Electronics 60.53%; Furniture 30.58%; Home 7.43%; Stationery 1.46%.
1.8 Peak day	2025-02-10 — $860.
2 Functions	Eva Martin $132; Diego Park $480; Cara Singh $40; Ben Ortiz $445; Alice Chen $45 (dates and elapsed days are shown in the screenshot).
3 NULLs	22 rows; 9 populated discounts; AVG discount 0.0722.
5.1 Mongo month	2025-01 $2,915; 2025-02 $1,436; 2025-03 $1,142.
5.2 Mongo top 5	Laptop $3,200; Standing Desk $1,680; Desk Lamp $300; Wireless Mouse $125; Ceramic Mug $108.
5.3 Mongo city	Austin $459 (3 orders); Boston $225.33 (3); Chicago $8 (1); Seattle $572 (6).

5. MongoDB aggregation pipelines

```
const orders = db.getSiblingDB("RetailSalesReportMongo").orders;
```

5.1 Month-wise sales

```
orders.aggregate([
  {$match: {orderDate: {$gte: ISODate("2025-01-01"), $lt: ISODate("2025-04-01")}}},
  {$unwind: "$items"},
  {$group: {_id: {$dateToString: {format: "%Y-%m", date: "$orderDate"}}, revenue: {$sum: {$multiply:
["$items.quantity", "$items.unitPrice"]}}}},
  {$sort: {_id: 1}}
]).toArray();
```

Output for pipeline 5.1

```
ubuntu@sandbox:~$ mongosh --quiet /home/ubuntu/retail_sales_set/mongo_monthly.js
5.1 Month-wise sales
[
  {
    _id: '2025-01',
    revenue: 2915
  },
  {
    _id: '2025-02',
    revenue: 1436
  },
  {
    _id: '2025-03',
    revenue: 1142
  }
]
ubuntu@sandbox:~$
```

5.2 Top 5 products by revenue

```
orders.aggregate([
  {$match: {orderDate: {$gte: ISODate("2025-01-01"), $lt: ISODate("2025-04-01")}}},
  {$unwind: "$items"},
  {$group: {_id: {id: "$items.productId", name: "$items.productName"}, revenue: {$sum: {$multiply: ["$items.quantity",
"$items.unitPrice"]}}}},
  {$sort: {revenue: -1, "_id.id": 1}}, {$limit: 5}
]).toArray();
```

Output

```
ubuntu@sandbox:~$ mongosh --quiet /home/ubuntu/retail_sales_set/mongo_top_products.js
5.2 Top 5 products by revenue
[
  {
    _id: {
      id: 201,
      name: 'Laptop'
    },
    revenue: 3200
  },
  {
    _id: {
      id: 204,
      name: 'Standing Desk'
    },
    revenue: 1680
  },
  {
    _id: {
      id: 206,
      name: 'Desk Lamp'
    },
    revenue: 300
  },
  {
    _id: {
      id: 202,
      name: 'Wireless Mouse'
    },
    revenue: 125
  },
  {
    _id: {
      id: 203,
      name: 'Ceramic Mug'
    },
    revenue: 108
  }
]
ubuntu@sandbox:~$
```

5.3 Average basket value by city

```
orders.aggregate([
  {$match: {orderDate: {$gte: ISODate("2025-01-01"), $lt: ISODate("2025-04-01")}}},
  {$addFields: {basketValue: {$sum: {$map: {input: "$items", as: "i", in: {$multiply: ["$$i.quantity",
"$$i.unitPrice"]}}}}}},
  {$group: {_id: "$customer.city", avgBasketValue: {$avg: "$basketValue"}, orders: {$sum: 1}}},
  {$sort: {_id: 1}}
]).toArray();
```

Output

